USN | | | | | | | | | | | [22BSMA22]

SECOND SEMESTER B.Tech EXAMINATION, JULY - 2023

PARTIAL DIFFERENTIAL EQUATIONS AND LAPLACE TRANSFORMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 X 2=10**
1.	**a.**	Evaluate $\int_0^2 \int_0^3 x^2 y dy dx$	L-2	CO1	**02**
	b.	Find the Laplace Transforms of $\sqrt{t}$.	L-2	CO3	**02**
	c.	Find $L^{-1}\left\{\frac{e^{-3s}}{(s-4)^2}\right\}$	L-2	CO4	**02**
	d.	Solve $x\,dy - y dx = 0$ by variable separable method.	L-3	CO5	**02**
	e.	Write the complementary function of the differential equation $(D^2 + p^2)X = 0$.	L-3	CO6	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Evaluate $\int_0^1 \int_x^{\sqrt{x}} (x^2 + y^2) dy\, dx$	L-3	CO1	**05**
		OR			
	b.	Evaluate $\iint_R x\,y\; dx\,dy$ where R is the region bounded by the co-ordinate axes and the line $x + y = 1$.	L-3	CO1	**05**
3.	**a.**	Find the Laplace Transform of $f(t) = \sin 5t \cos 2t$	L-3	CO3	**05**
		OR			
	b.	Find the Laplace Transform of $(3t^2 + 4t + 5)u(t-3)$.	L-3	CO3	**05**
4.	**a.**	Find the inverse Laplace transform of $\frac{3s+1}{(s+1)^4}$	L-3	CO4	**05**
		OR			
	b.	Find $L^{-1}\left[\frac{s}{(s^2+a^2)^2}\right]$	L-3	CO4	**05**

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5.	a.	Solve $\frac{\partial^3 z}{\partial x^2 \partial y} = \cos(2x+3y)$ by direct integration	L-3	CO5	05
		OR			
	b.	Solve $y^2 zp = x^2(zq+y)$.	L-3	CO5	05
6.	a.	Solve by the method of separation of variables $4\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 3u$	L-3	CO6	05
		OR			
	b.	Write Matlab code to evaluation the double integral by change of order of integration.	L-3	CO1	05
7.	a.	Evaluate $\int_0^1 \int_{x^2}^{2-x} xy\, dy\, dx$ by changing the order.	L-3	CO1	05
		OR			
	b.	Find the Laplace transform of $\frac{2\sin t \sin 5t}{t}$	L-3	CO3	05
		PART-C	**6 X 10=60**		
8.	a.	Find by double integration the area enclosed by the curve $r = a(1+\cos\theta)$ between $\theta = 0$ and $\theta = \pi$.	L-3	CO2	10
		OR			
	b.	A pyramid is bounded by three co-ordinate planes and the plane $x+2y+3z=6$. Compute the volume by double integration	L-3	CO2	10
9.	a.	Express $f(t) = \begin{cases} \cos t, 0 < t < \pi \\ \cos 2t, \pi < t > 2\pi \\ \cos 3t, t > 2\pi \end{cases}$ in terms of unit step function and hence find its Laplace transform.	L-3	CO3	10
		OR			
	b.	Find the Laplace transform of the periodic function $f(t) = \begin{cases} E\sin\omega t, & 0 < t \le \frac{\pi}{\omega} \\ 0, & \frac{\pi}{\omega} < t \le \frac{2\pi}{\omega} \end{cases}$.	L-3	CO3	=10
10.	a.	Verify convolution theorem for the functions $f(t) = t$ and $g(t) = \cos t$	L-3	CO4	10
		OR			
	b.	Solve the differential equation $y'' + 4y' + 3y = e^t, y(0) = 1 = y'(0)$.	L-3	CO4	10

11.	**a.**	Solve $\frac{\partial^3 u}{\partial x \partial t} = e^{-t} cosx$ by direct integration, given that $u = 0$ when $t = 0$ *and* $\frac{\partial u}{\partial t} = 0$ *at* $x = 0$.	L-3	CO5	**10**
		OR			
	b.	Solve $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} = 0$, by the method of separation of variables.	L-3	CO5	**10**
12.	**a.**	Find various possible solutions of one-dimensional wave equation by the method of separation of variables	L-3	CO6	**10**
		OR			
	b.	Obtain the D'Alembert's solution of the one dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ given that $u(x,0) = f(x) = l^2 - x^2$ and $u_t(x,0) = 0$.	L-3	CO6	**10**
13.	**a.**	Write the Matlab code for evaluation for triple integral.	L-3	CO2	**10**
		OR			
	b.	Find the inverse Laplace transform of the function $\frac{s}{(s^2+a^2)^2}$ using convolution theorem.	L-3	CO4	**10**